AI-Powered Tour Guide Booking System with Voice and RAG

25-26J-150

September 2025

SLIIT
FACULTY OF COMPUTING

AI-Powered Tour Guide Booking System with Voice and RAG

P.R.I.PRAVEAN
IT 21 191 060

IT Specializing in Information Systems Engineering

srilanka

Research Problem

- Tourists face difficulty in finding trusted, verified guides
- Poor personalization and static recommendations don't consider preferences, or real-time data.

Specific Objectives

- Develop a secure AI-powered web application for tour guide bookings that integrates multi-level document verification.
- Implement an intelligent matching and recommendation engine using Retrieval-Augmented Generation (RAG) to provide personalized, and context-aware guidance for tourist

Sub Objectives

- Design user interfaces and workflows for tourist voice queries, guide registration uploads booking inputs.
- Integrate fraud detection and multilingual chat agent features to enhance trust and improve real-time tourist-guide interactions.

Functional Requirement

- Multi Document Verification – Validation of NIC, passport, licenses, and police check to categorize guides as verified or non-verified.
- Tourist Guide Matching – AI-powered RAG-based engine to recommend guides based on tourist preferences, location, and real-time factors.

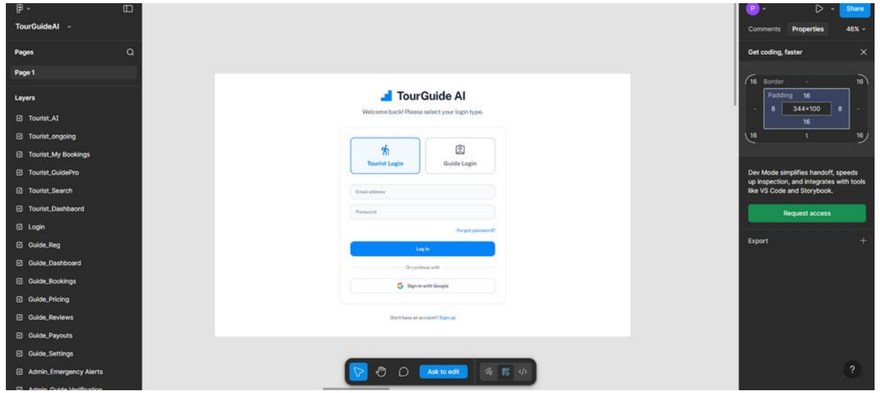
Non-Functional Requirement

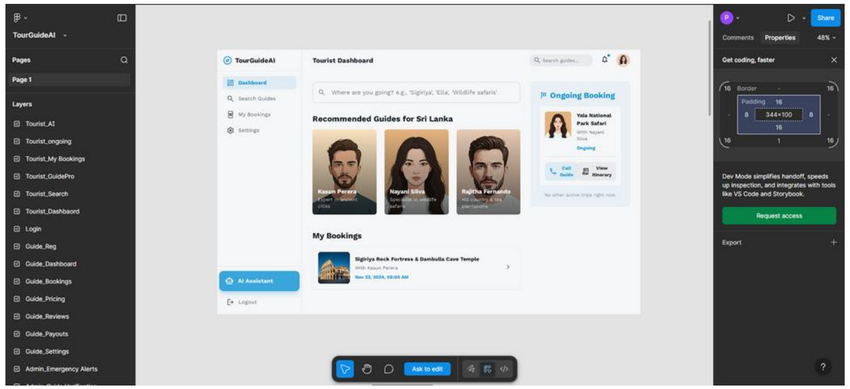
- Security
- Scalability
- Performance
- Reliability
- Cross-Platform Accessibility
- Usability

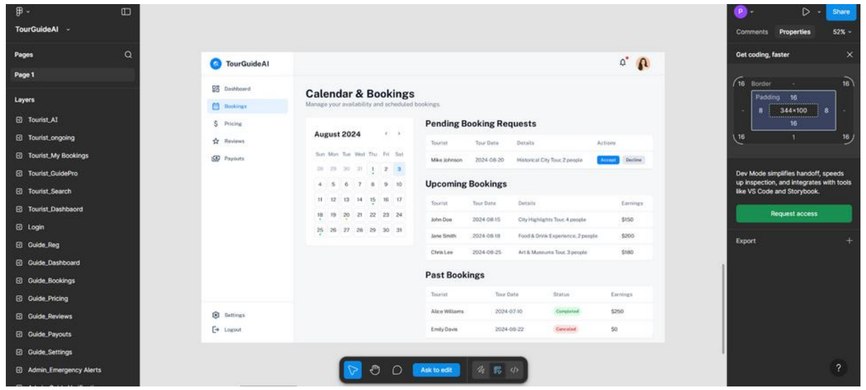
User Requirements addressed by the solution

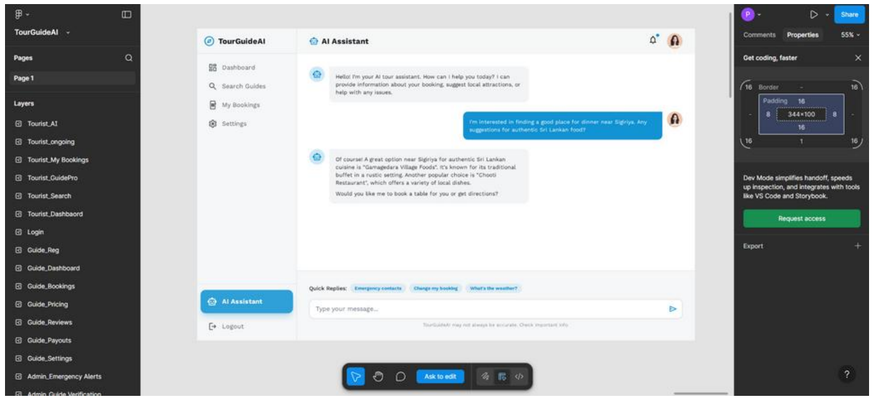
- Admin: Tour guides accept or reject the Tour guides
- Tour guides **:** NIC, Police Report and SL Tourism card to upload and accept the trip
- Tourist : Assign the trip in the system by agent.

•Design Excellence/Contribution





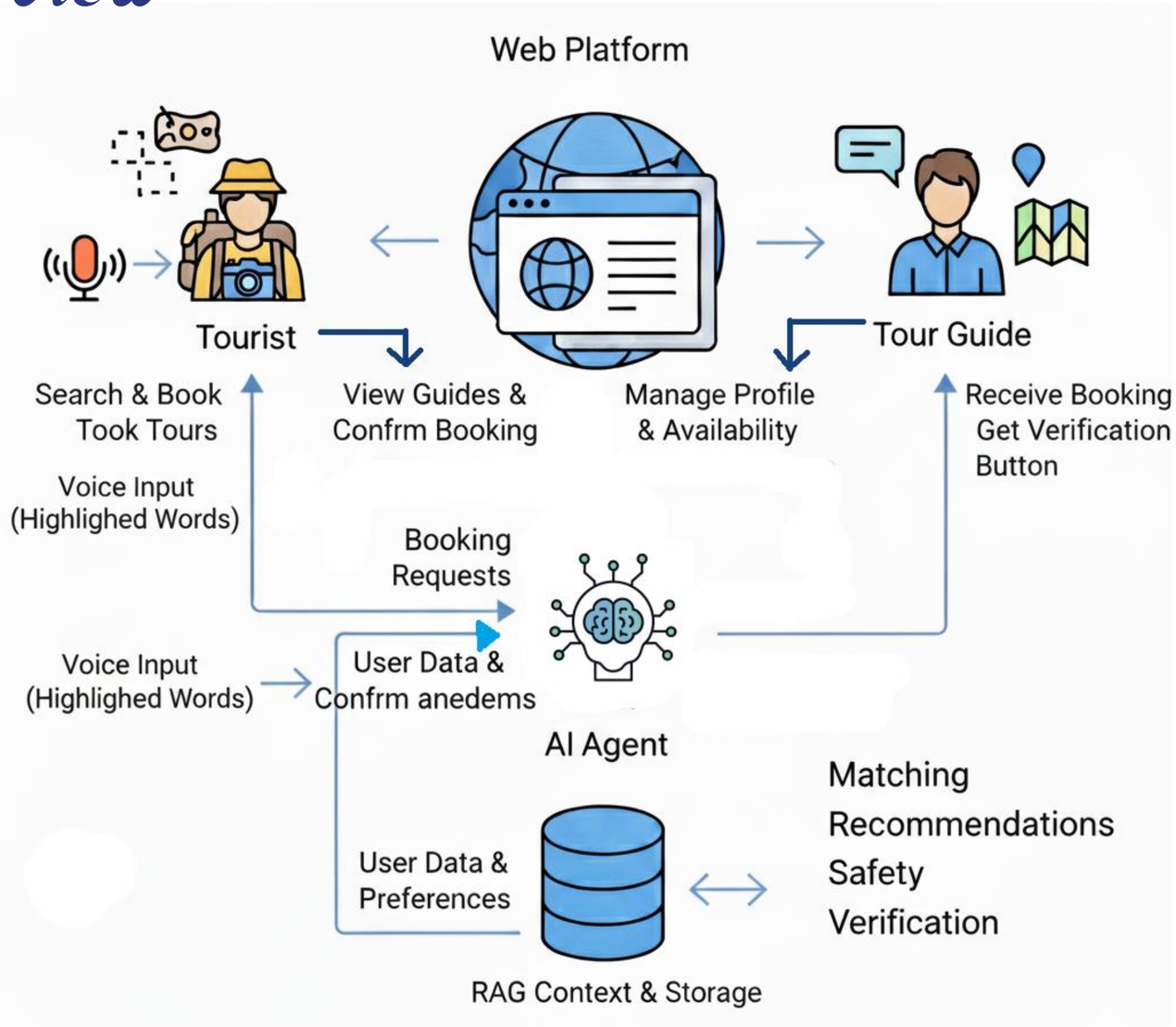




Technologies

- Frontend: React.js for interactive UI.
- Backend: Node.js/Express with secure API integrations and Python.
- Database: Vector DB for RAG retrieval + firebase DB.
- AI/ML: NLP, keyword extraction, RAG for intelligent responses.
- Multilingual Support: Sinhala, Tamil, English chat agent voice interface.

System Overview



Commercialization

- Commission-based revenue from bookings.
- Subscription plans for verified guides (premium listings).

Target Audience and Market Size

- Primary: local tourists seeking and verified guides.
- Secondary: Professional tour guides seeking visibility and trust.
- Tertiary: Tourism boards, hotels, and travel agencies.

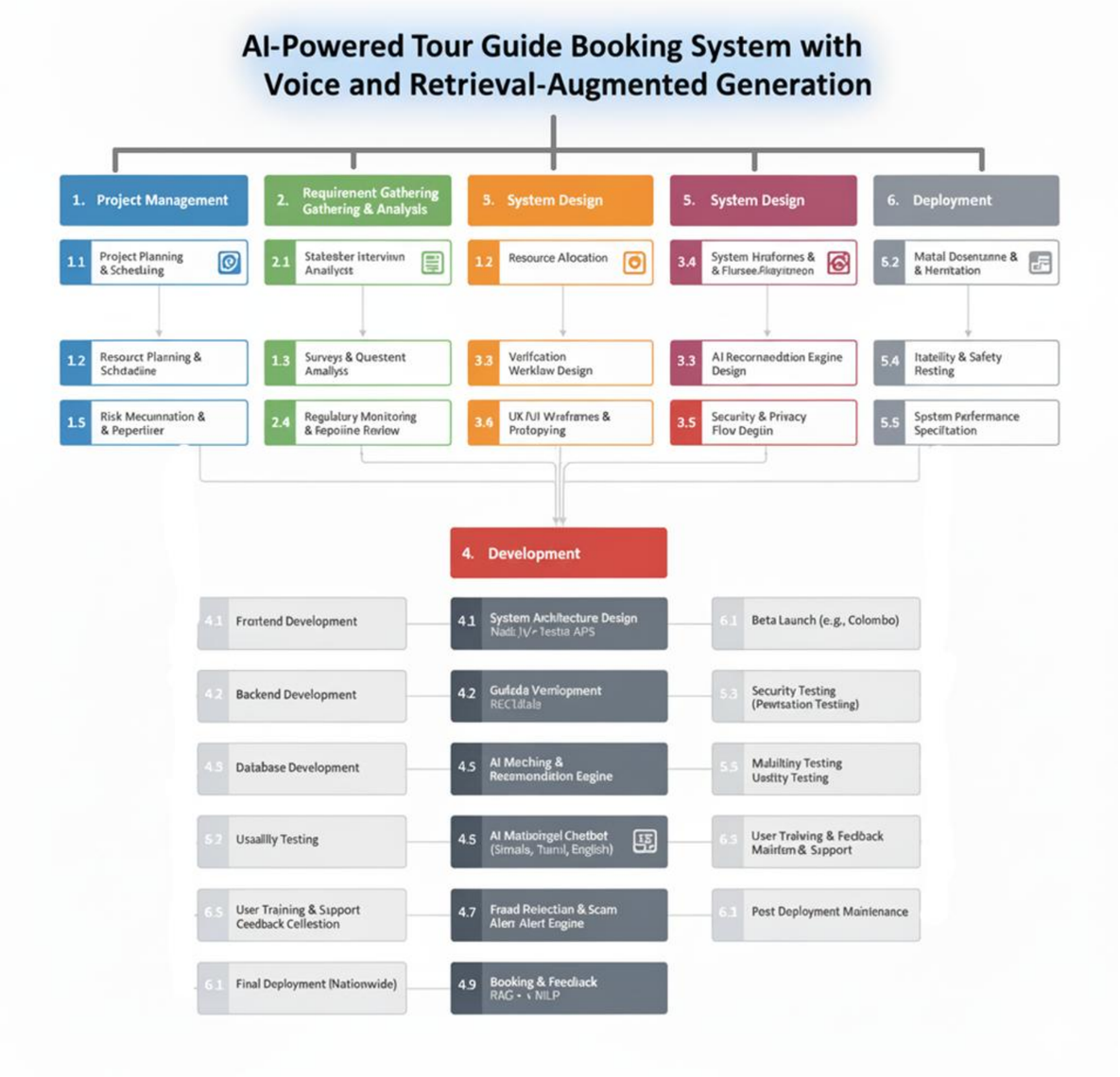
Investment / Cost & Recovery

- Commission fees (10–15%) per booking.
- Subscription fees from guides.
- Sponsored placements from travel partners.

What is Unique in Your Solution

- For Tourists: Safety, trust, AI support, personalized journeys.
- For Guides: Verified exposure, fair marketplace, protection from fraud.
- For Sri Lanka: Enhanced tourism reputation, repeat visitors, digital innovation leadership.

WBS


AI-Powered Tour Guide Booking System with Voice and Retrieval-Augmented Generation
1. Project Management
2. Requirement Gathering Gathering & Analysis
3. System Design
5. System Design
6. Deployment
1.1 Project Planning & Scheduling
2.1 Stakeholder Interviews Analysis
1.2 Resource Allocation
3.4 System Hardware & Software Requirements
5.2 Manual Documentation & Verification
1.2 Resource Planning & Scheduling
1.3 Surveys & Questionnaire Analysis
3.3 Verification Workflow Design
3.3 AI Recommendation Engine Design
5.4 Usability & Safety Testing
1.5 Risk Mitigation & Reporting
2.4 Regulatory Monitoring & Compliance Review
3.6 UX/UI Wireframes & Prototyping
3.5 Security & Privacy Flow Design
5.5 System Performance Specification
4. Development
4.1 Frontend Development
4.1 System Architecture Design
6.1 Beta Launch (e.g., Colombo)
4.2 Backend Development
4.2 Guide Development
5.3 Security Testing (Penetration Testing)
4.3 Database Development
4.5 AI Matching & Recommendation Engine
5.5 Multilingual Testing Usability Testing
5.2 Usability Testing
4.5 AI Multilingual Chatbot (Sinhala, Tamil, English)
6.5 User Training & Feedback Mechanism & Support
6.5 User Training & Support Feedback Collection
4.7 Fraud Detection & Scam Alert Alert Engine
6.1 Post Deployment Maintenance
6.1 Final Deployment (Nationwide)
4.9 Booking & Feedback RAG + NLP

References

- X. Zhou, Z. Zhou and X. Shi, "FCH-TTS: Fast, Controllable and High-quality Non-Autoregressive Text-to-Speech Synthesis," 2022 International Joint Conference on Neural Networks (IJCNN), Padua, Italy, 2022, pp. 1-8.
- S. Mohanty and B. K. Swain, "S2S system for voice oriented tourism information delivery in Indian context," 2013 International Conference Oriental COCOSDA held jointly with 2013 Conference on Asian Spoken Language Research and Evaluation (O-COCOSDA/CASLRE), Gurgaon, India, 2013, pp. 1-5.
- G. Boza-Quispe, J. Montalvan-Figueroa, J. Rosales-Huamaní and F. Puente-Mansilla, "A friendly speech user interface based on Google cloud platform to access a tourism semantic website," 2017 CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON), Pucon, Chile, 2017, pp. 1-6.

Thank You